The witness game beyond Erdős Problem 848: higher powers, $ab + c$, and the third density

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Abstract

Erdős Problem 848 asks for the largest $A \subseteq \{1, \dots, N\}$ with $ab + 1$ never squarefree. The answer is the class 7 mod 25. In a companion paper [M] we proved this for every N by a finite "witness game" on residue types. That paper also determined the second-best density, 1/50, attained by a two-class set F. Here we test how far the method reaches.

- **Higher powers.** For $ab + 1$ never k-free, the maximum is a class mod 5^k for every $k \geq 3$ and all large N, with an explicit threshold. For $3 \leq k \leq 100$ this holds for every N, with exactly two small exceptions, at $k = 3$ and $k = 4$. For cubes the maximisers are listed for every N.
- **Other forms.** For $ab + c$ never squarefree, the maximum for large N is a root class of $x^2 + c \equiv 0 \pmod{p^2}$. This holds for 49 squarefree $c \leq 100$, and for 33 of them the second-best density is also determined. The counting lemmas needed for general c are proved here.
- **The third density for 848.** The densest valid set not inside a class, F or F's mirror has density between $1/25 - 1/(4\pi^2) = 0.014670$ and 0.014910. The lower bound is one class plus one outsider, and we conjecture that it is sharp.

The computations use exact clique search and CP-SAT with rounded-up objective coefficients. All code is public.

1. Introduction

Call $A \subseteq [N]$ *valid* if $ab + 1$ is not squarefree for all $a, b \in A$, including $a = b$. The class $C_7 = \{n \equiv 7 \pmod{25}\}$ is valid, since $7^2 + 1 = 50$. In [M] we proved that it is the largest valid set for every N, together with stability and rigidity results and the second-best density.

The method of [M] has three ingredients:

- a *self-forcing class*: a residue class in which $ab + 1$ is always divisible by a fixed square;
- a *positive local gap*: a finite game on residue types showing that any valid set not inside such a class is smaller by a constant factor;
- *control of large primes*: counting lemmas showing that squares p^2 with p large contribute $o(N)$, uniformly in the witness.

This paper asks how special these are to $x^2 + 1$ and to squares. Section 2 proves the counting lemmas for $ab + c$ with a general squarefree $c \geq 1$. Section 3 uses them, with $c = 1$, to bracket the third density of 848. Section 4 treats higher powers, and Section 5 the forms $ab + c$.

We use the notation of [M] throughout. A *type* is a set of residue classes modulo a fixed modulus L. Type s *forces* type t if some fixed square divides $ab+c$ for every $a \in t$ and $b \in s$. A type is *self-forcing* if it forces itself. For a set A, P is the set of types it meets, its *core* K is the set of types in P forced by every type in P, and $F(K)$ is the set of types forcing every member of K. Forcing is symmetric, so $P \subseteq F(K)$.

2. Counting lemmas for $ab + c$

Fix a squarefree $c \geq 1$ and a modulus L. The *layer primes* are the primes p whose squares are built into the types, through $L = 36 \cdot \prod p^2$ or similar. Every other prime q is a *luck prime*. For a luck prime q, the event $q^2 \mid ab + c$ is not decided by the types and is counted directly.

Lemma 2.1 (large luck primes). Let $1 \leq b \leq N$ and $Y \geq c$. Then

$$\#\{a \leq N : q^2 \mid ab + c \text{ for some prime } q > Y\} \leq c \cdot 2^{\omega(b)+\omega(c)+1} \cdot \left(2(N + c)/Y + (N^2 + c)/Y^2\right).$$

Proof. Let $q > Y$ have $q^2 \mid ab + c$. Then $q \nmid c$, since $q > c$, and so $q \nmid b$. Write $ab + c = mq^2$, so that $1 \leq m \leq M = (bN + c)/Y^2$ and $mq^2 \equiv c \pmod{b}$.

Count the units s mod b with $ms^2 \equiv c \pmod{b}$ prime power by prime power. Let $\ell^e \parallel b$.

- If $\ell \nmid c$, then $\ell \nmid m$, and $s^2 \equiv cm^{-1}$ has at most 2 solutions mod ℓ^e (at most 4 if $\ell = 2$).
- If $\ell \mid c$, then $\ell \parallel c$ because c is squarefree, and $\ell \mid m$. For $e = 1$ every unit works, which gives fewer than ℓ solutions. For $e \geq 2$, write $m = \ell m'$ and $c = \ell c'$. Then $m's^2 \equiv c' \pmod{\ell^{e-1}}$ with c' a unit, which has at most 4 solutions mod ℓ^{e-1}, so at most 4ℓ mod ℓ^e.

So q lies in at most $c \cdot 2^{\omega(b)+\omega(c)+1}$ classes mod b. Since $q \leq \sqrt{(bN + c)/m}$, each m contributes at most $c \cdot 2^{\omega(b)+\omega(c)+1} \left(\sqrt{(bN + c)/m}/b + 1\right)$. Summing over $m \leq M$ with $\sum m^{-1/2} \leq 2\sqrt{M}$ gives $c \cdot 2^{\omega(b)+\omega(c)+1} \left(2(bN + c)/(bY) + M\right)$. This is at most the bound stated, since $1 \leq b \leq N$. ■

For $c = 1$ this is Lemma 8.3 of [M]. With $Y = N^{0.6}$ it is $N^{0.8+o(1)}$, uniformly in $b \leq N$, because $2^{\omega(b)} = N^{o(1)}$.

Lemma 2.2 (one witness). Let t be a type of density w_t and let $b \leq N$. For each luck prime q, let $d_q(t, b)$ be the relative density in t of the a with $q^2 \mid ab + c$. This density is determined by the classes of b mod L and mod q^2. Then, uniformly in $b \leq N$,

$$\#\{a \in t \cap [N] : q^2 \mid ab + c \text{ for some luck prime } q\} \leq w_t \cdot \left(1 - \prod_q (1 - d_q(t, b))\right) \cdot N + o(N),$$

and also $\leq w_t \cdot \sum_q d_q(t, b) \cdot N + o(N)$.

The densities $d_q(t, b)$ are as follows.

- For a luck prime $q \nmid cL$:
  - if $q \nmid b$, a must lie in the single class $-cb^{-1}$ mod q^2, so $d_q = 1/q^2$;
  - if $q \mid b$, then $ab + c \equiv c \not\equiv 0 \pmod{q}$, so $d_q = 0$.
- For a luck prime $q \mid c$, the type records whether $q \mid a$. Then $q^2 \mid ab + c$ needs q to divide exactly one of a and b.

  - If $q \mid b$ and $q \nmid a$, then a lies in one nonzero class mod q, so $d_q = 1/(q-1)$.
  - If $q \mid a$ and $q \nmid b$, then a/q lies in one class mod q, so $d_q = 1/q$.
  - In the other two cases, $d_q = 0$.
- For a luck prime q whose root classes are recorded in the types (as in Sections 3 and 5):
  - if a lies in a root class R and b does not, then $d_q = 0$, because $ab \equiv -c \pmod{q^2}$ would put b in R;
  - if a lies in no root class, then $-cb^{-1}$ is one of the q^2-2 remaining classes, so $d_q \le 1/(q^2-2)$.

Proof. Fix Q. For the luck primes $q \le Q$, the events are conditions on a modulo $L \cdot \prod_{q \le Q} q^2$. By the Chinese remainder theorem, the count is exact up to $O(1)$ per class, where the implied constant depends on Q. This gives the product for $q \le Q$, and the union bound gives the sum.

For $Q < q \le Y$ with $Y = N^{0.6}$, each q contributes at most $N/q^2 + 1$, or $N/q + 1$ when $q \mid c$; but $q \mid c$ means $q \le c \le Q$ once $Q \ge c$. So these primes contribute at most $N/Q + \pi(Y)$. For $q > Y$ use Lemma 2.1. Letting $Q \to \infty$ slowly with N, the total error is $o(N)$, uniformly in b. ■

Lemma 2.3 (large primes in $x^2 + c$). For every $Q \ge 1$,

$$\#\{x \le N : p^2 \mid x^2+c \text{ for some prime } p > Q\} \le (2/Q)\cdot N + 2\pi\left(\sqrt{N^2+c}\right) = (2/Q)\cdot N + O(N/\log N).$$

Proof. If $p \mid c$ then $p^2 \nmid x^2 + c$: $p \mid x$ would give $p^2 \mid x^2$, and c is squarefree. So $p \nmid c$, and $x^2 \equiv -c \pmod{p^2}$ has at most two roots. Each root class meets $[N]$ in at most $N/p^2 + 1$ elements, and $p \le \sqrt{N^2+c}$. Sum over $p > Q$, using $\sum_{p>Q} 1/p^2 \le 1/Q$. ■

For $c = 1$, [M] used Pell's equation to get a power saving here. For an asymptotic statement this cruder bound suffices, and it holds for every c.

Lemma 2.4 (joint estimate with the own witness). Let t be a type that is not self-forcing, and let $b \in t$. Write $V_c = \{x : x^2 + c \text{ not squarefree}\}$, and ρ_p for the number of roots of $x^2 \equiv -c \pmod{p^2}$. Suppose t records, for each luck prime $q \mid c$, whether $q \mid a$. Then, uniformly in $b \le N$,

$$\#\{a \in t \cap [N] : a \in V_c,\ ab + c \text{ not squarefree}\} \le w_t \cdot (\alpha_t \cdot \beta_t^0 + \kappa_t) \cdot N + o(N),$$

where

- α_t is the relative density in t of the a that have $p^2 \mid a^2 + c$ for some luck prime p, bounded by the union bound $\sum \rho_p/p^2$ (a luck prime whose root classes are recorded contributes nothing here, since t is not self-forcing and so lies outside them);
- $\beta_t^0 = \sum d_q(t, b)$ over luck primes $q \nmid c$;
- $\kappa_t = \sum 1/p^2$ over the luck primes p with $\rho_p = 2$.

Proof. Since t is not self-forcing and $b \in t$, the layer primes force nothing between a and b. So $a \in V_c$ needs $p^2 \mid a^2 + c$ for some luck prime p, and $ab + c$ not squarefree needs $q^2 \mid ab + c$ for some luck prime q.

A luck prime $q \mid c$ contributes nothing. The elements a and b lie in the same type, so q divides both or neither. If q divides both, $ab + c \equiv c \pmod{q^2}$ and $q \parallel c$. If q divides neither, $ab + c \equiv ab \not\equiv 0 \pmod{q}$.

Fix Q and consider $p, q \le Q$. For $p \ne q$, the Chinese remainder theorem gives the product of the densities. For $p = q$, a must lie in a root class mod p^2 and in the class $-cb^{-1}$ mod p^2. The two root

classes are distinct, so this leaves at most one class, and contributes $1/p^2$. Up to the CRT errors, this gives $w_t(\alpha_t\beta_t^0 + \kappa_t)N$.

For $p > Q$, drop the condition on q and use Lemma 2.3. For $q > Q$, drop the condition on p and use the tail of Lemma 2.2. Let $Q \to \infty$. ■

Lemma 2.5 (core reduction). Let $A \subseteq [N]$ be c-valid, meaning $ab + c$ is not squarefree for all $a, b \in A$. Let P be the set of types A meets, with core K. Suppose that for each type t there is a constant b_t with these properties:

- if t is self-forcing, b_t is at least the luck density of Lemma 2.2 for every witness that does not force t;
- otherwise, b_t is at least the constant of Lemma 2.4.

Then

$$|A| \le U(K) \cdot N + o(N), \quad \text{where } U(K) = w(K) + \sum_{t \in F(K) \setminus K} b_t \cdot w_t.$$

Proof. As in [M, Lemma 8.5]. Every element of A is compatible with every other. A self-forcing type $t \in P \setminus K$ is not forced by some $s \in P$, so any element of $A \cap s$ is a witness for Lemma 2.2. A type that is not self-forcing uses an element of itself as the witness for Lemma 2.4. Core types count in full. Since $P \subseteq F(K)$, summing over $t \in P$ gives the bound. The number of types is fixed, so the $o(N)$ terms add up to $o(N)$. ■

Maximising $U(K)$ over the cores that a given family of sets can have is a finite problem. We solve it with CP-SAT [OR-tools], with every objective coefficient rounded up to a multiple of 10^{-9}, so the reported optimum is an upper bound.

The variables are as follows. A *slab* is a residue class for the layers other than 4 and 9, together with the root-class memberships described below. A *cell* is a residue mod 36, together with the divisibility flags. Then K is given by a closed set of cells in each slab, and $F(K)$ by its intersections, exactly as in [M, Section 8.4]. The code is `coregame.py`.

3. The third density of Problem 848

Let $F = \{7 \bmod 25, 1 \bmod 4\} \cup \{18 \bmod 25, 3 \bmod 4\}$ and let F' be its mirror image. In [M] we showed that every valid set not inside C_7 or C_{18} has at most $N/50 + O(N^{0.81})$ elements, and that F and F' attain this. We also showed that for $N \ge 10^{23.6}$, a valid set not inside a class with at least $(1/50 - 0.0008)N$ elements lies in F or F'. Let $\lambda_3(N)$ be the largest size of a valid $A \subseteq [N]$ contained in none of C_7, C_{18}, F and F'.

Theorem 3.1. 1. $\liminf \lambda_3(N)/N \ge 1/25 - 1/(4\pi^2) = 0.0146697\ldots$. 2. $\limsup \lambda_3(N)/N \le 0.014910$.

So for all large N, a valid set with more than $0.01491N$ elements lies inside C_7, C_{18}, F or F'. This sharpens the density $1/50 - 0.0008 = 0.0192$ of [M, Theorem 8.9] to 0.01491, though without an explicit threshold for N.

Proof of (1). Take a prime $x > 3$ with $x \equiv 18 \pmod{25}$ and $x \equiv 3 \pmod 4$, and let

$$A = \{x\} \cup \{a \in C_7 \cap [N] : ax + 1 \text{ not squarefree}\}.$$

Pairs inside C_7 are forced by 25. Also $x^2+1 \equiv 0 \pmod{25}$, and pairs with x hold by definition, so A is valid. It contains $x \in C_{18}$ and elements of C_7 that are $\equiv 1$ and $\equiv 3 \pmod 4$, so it lies in none of the four sets.

For $a \in C_7$ we have $ax+1 \equiv 2 \pmod{25}$, so 25 never divides $ax+1$. Among the $a \in C_7$:

- 4 divides $ax+1$ for one class of a mod 4;
- 9 divides it for one class mod 9;
- p^2 divides it for one class mod p^2, when $p \geq 7$ and $p \neq x$;
- x^2 never divides it.

So the $a \in C_7$ with $ax+1$ squarefree have density

$$(1/25) \cdot (3/4) \cdot (8/9) \cdot \prod_{p \geq 7,\, p \neq x} (1-p^{-2}) = (1/(4\pi^2)) \cdot (1-x^{-2})^{-1},$$

using $\prod_{p \geq 7}(1-p^{-2}) = (6/\pi^2) \cdot (4/3) \cdot (9/8) \cdot (25/24) = 75/(8\pi^2)$. Hence $|A|/N \to 1/25 - (1/(4\pi^2))(1-x^{-2})^{-1}$ as $N \to \infty$, and letting $x \to \infty$ gives (1). The script `lam3_lower.py` checks the count: 0.014671 at $N = 2 \cdot 10^7$ with $x = 100043$. ■

This is the "one class plus one outsider" configuration. The outsider forces exactly a third of C_7, through 4 and 9 [M, Lemma 2.2]. The other two thirds survive only where $ax+1$ picks up a square factor by chance. Its density is the sharp constant g_1^* of [M, Theorem C(1)].

Proof of (2). Apply Lemma 2.5 with $c = 1$ and the following types:

- the class mod 25;
- the cell mod 36;
- membership in the classes $S_p^{\pm} = \{a \equiv \pm \iota_p \pmod{p^2}\}$ for $p = 13$ and 17, where $\iota_p^2 \equiv -1$.

Two elements of the same S_p^{σ} are forced. An element of S_p^{σ} and one outside it never have $p^2 \mid ab+1$, since $ab \equiv -1 \pmod{p^2}$ would put the second element in S_p^{σ} too. Inside the complement of $S_p^{\pm}$, the prime p acts as a luck prime with density $1/(p^2-2)$. So the luck primes are $q \geq 7$, and the types outside $C_7 \cup C_{18} \cup S_{13}^{\pm} \cup S_{17}^{\pm}$ are in V only through primes $p \geq 29$.

The constants are as follows.

- For self-forcing types, $b_t = 1 - \prod(1-q^{-2})$, with the product over $7 \leq q \leq 10^7$ and the factor for $p = 13, 17$ replaced by $1 - 1/(p^2-2)$, plus $\sum_{q > 10^7} q^{-2}$. This is 0.0502.
- For the remaining types, b_t is the Lemma 2.4 constant, 0.0050. It was 0.0158 for the unrefined R-types of [M].

The admissible cores are those whose $F(K)$ has a type outside each of C_7, C_{18}, F and F'. Maximising U over them is a problem on 225 slabs × 21 cells. `third_cp.py` solves it to optimality and finds 0.0149093. ■

The optimal core is the class-plus-outsider core: twelve cells of one class, all forced by a single cell. So the game identifies the right configuration, and the gap of 0.00024 to the construction has two sources.

- Most of it comes from elements of the outsider's cell, in the other classes mod 25, that lie in V through primes $p \geq 29$. This part shrinks as more classes S_p are split off: the bound is 0.0149645 with S_{13} alone.

- The rest, $\beta/900 \approx 0.000056$, is the outsider cell itself. The game charges it at rate β, although the construction has a single element there.

Closing the second part needs a bound for pairs of sets X, Y with $xy+1$ never squarefree that is better than one witness on each side.

The unrefined enumeration of [M, Section 8.4] gives only 0.0186 here (`third.py`). In that version the R-types carry the full constant $\kappa = \sum_{p\equiv 1(4),\, p\geq 13} p^{-2} = 0.0138$ from the coincidence $p = q$, which the refinement removes.

Conjecture 3.2. $\lambda_3(N)/N \to 1/25 - 1/(4\pi^2)$.

4. Higher powers

4.1 Cubes

Call $A \subseteq [N]$ *cube-valid* if $ab+1$ is not cube-free for all $a, b \in A$, including $a = b$. Write $V_3 = \{x : x^2+1 \text{ not cube-free}\}$. Since $57^2+1 = 2\cdot 5^3 \cdot 13$, the classes 57 and 68 mod 125 are cube-valid, playing the role of 7 and 18 mod 25. Write $c_{57}(N) = \#\{n \leq N : n \equiv 57 \pmod{125}\}$.

Theorem 4.1. For every N, every cube-valid $A \subseteq [N]$ has $|A| \leq c_{57}(N)$. Equality holds exactly in the following cases:

- $A = C_{57} \cap [N]$;
- $A = C_{68} \cap [N]$, when $c_{68}(N) = c_{57}(N)$;
- for $193 \leq N \leq 306$, also $A = \{182, 193\}$;
- for $239 \leq N \leq 306$, also $A = \{57, 239\}$ and $A = \{193, 239\}$.

The exceptional sets work through 8 and 27: $182\cdot 193+1 = 3^3\cdot 1301$, $57\cdot 239+1 = 2^3\cdot 13\cdot 131$ and $193\cdot 239+1 = 2^4\cdot 3\cdot 31^2$. Locally the picture is the same as for 848. The densest congruence-defined cube-valid set not inside one class is $\{57 \bmod 125, 1 \bmod 8\} \cup \{68 \bmod 125, 7 \bmod 8\}$, of density $1/500$ (`wgame4.py`).

The proof is much shorter than for 848, for two reasons. The local gap is far larger, so a single witness suffices and the full game is not needed. And a cube $q^3 > N^2+1$ can never divide $ab+1$, so no lattice lemma is needed either.

Lemma 4.2 (one witness). Let A be cube-valid, and suppose some $b \in A$ lies outside C_{57}. Then

$$|A \cap C_{57}| \leq (N/125)\cdot(1/8 + 1/27 + \beta_3) + 2 + \pi((N^2+1)^{1/3}), \quad \text{where } \beta_3 = \sum_{q\geq 7} q^{-3} \leq 0.004726,$$

and the same holds for C_{68}.

Proof. Since $b \notin C_{57}$, 125 never divides $ab+1$ for $a \in C_{57}$. So some prime $q \neq 5$ has $q^3 \mid ab+1$, which puts a in one class mod q^3 (if $q \nmid b$). For $q = 2$ and $q = 3$ this is at most a fraction $1/8$ or $1/27$ of C_{57}, at a cost of 1 each. For $q \geq 7$, the class mod $125q^3$ contains at most $N/(125q^3)+1$ elements up to N. Also $q^3 \leq ab+1 \leq N^2+1$. ■

Proof of Theorem 4.1. Let A be cube-valid and inside neither C_{57} nor C_{68}. Then A has elements outside each class, so Lemma 4.2 applies to both. The remaining elements lie in V_3 outside $C_{57} \cup C_{68}$, so each has $p^3 \mid x^2+1$ for some $p \equiv 1 \pmod 4$ with $p \geq 13$. There are at most $2N\sigma_3 + 2\pi((N^2+1)^{1/3})$ of them, where $\sigma_3 = \sum p^{-3}$ over primes $p \equiv 1 \pmod 4$ with $p \geq 13$, so $\sigma_3 \leq 0.000756$.

- **Large N.** The total main term is $0.004178N$ against $c_{57}(N) > (N-57)/125 \approx 0.008N$. Checking cells of ratio 1.01 with exact prime counts to 10^8 and Rosser–Schoenfeld above (`cube_analytic.py`) gives $|A| < c_{57}(N)$ for $1.94 \cdot 10^6 \le N \le 10^{40}$. Beyond 10^{40} the error term $4\pi((N^2+1)^{1/3}) + 4$ is far below $(0.008 - 0.004178)N$.
- $700 \le N \le 2 \cdot 10^6$**.** Here `cubeM.c` computes the bound exactly. It finds the maximum over witnesses $b \in V_3 \setminus C_{57}$ of $\#\{a \in C_{57} \cap [N] : ab+1 \text{ not cube-free}\}$, likewise for C_{68}, and adds $|V_3 \cap [N]| - c_{57}(N) - c_{68}(N)$. All three are nondecreasing in N. On 20 cells of ratio 1.5, the bound at the top of each cell stays below c_{57} at the bottom of the cell. At its closest it reaches 89% of c_{57} (17 against 19 on $[2362, 3543]$). A brute-force recomputation by factorisation agrees at $N = 3000$.
- $N \le 700$**.** `cube_maxsets.py` lists all maximum cube-valid sets for every N by an exact clique search. An independent clique computation confirms the maximum $c_{57}(N)$ up to $N = 40{,}000$. ■

For squares the large-prime events were the main difficulty. For cubes all three ingredients are easy.

4.2 General k

Call $A \subseteq [N]$ *k-valid* if $ab+1$ is never k-free, for all $a, b \in A$ including $a = b$. Let $q = 5^k$, let r_k be the smaller root of $x^2 \equiv -1 \pmod q$, and write $c_k(N) = \#\{n \le N : n \equiv r_k \pmod q\}$. For example, $r_3 = 57$, $r_4 = 182$, $r_5 = r_6 = 1068$ and $r_7 = 32318$.

Theorem 4.3. 1. For every $k \ge 3$ there is an explicit N_k such that for $N \ge N_k$, every k-valid $A \subseteq [N]$ not inside one of the classes $\pm r_k \pmod q$ has fewer than $c_k(N)$ elements. So for $N \ge N_k$ the maximum is $c_k(N)$, and only the full classes attain it. Here $N_k/5^k$ is about 15,400 for $k = 3$, 555 for $k = 4$, at most 214 for $k \ge 5$, and at most 60 for $k \ge 40$. 2. For $3 \le k \le 100$ the maximum is $c_k(N)$ for every N, and only the full classes attain it, with exactly two exceptions. For $k = 3$ and $193 \le N \le 306$, the mixed sets of Theorem 4.1 also attain it. For $k = 4$ and $239 \le N \le 806$, the set $\{239\}$ also attains the maximum 1, because $239^2 + 1 = 2 \cdot 13^4$.

Proof of part 1. Lemma 4.2 holds with 2^k, 3^k and p^k in place of 8, 27 and p^3. A witness outside C_{r_k} forces at most a fraction $Z_k = 2^{-k} + 3^{-k} + \sum_{p \ge 7} p^{-k}$ of that class, plus one element for each prime p with $p^k \le N^2 + 1$. The rest of V_k has at most $2N\sigma_k + 2\pi((N^2+1)^{1/k})$ elements, where $\sigma_k = \sum p^{-k}$ over primes $p \equiv 1 \pmod 4$ with $p \ge 13$. The main term $2Z_k + 2q\sigma_k$ is largest at $k = 3$, where it is 0.52, and it falls to 0.21 at $k = 4$ and 0.09 at $k = 5$.

For $3 \le k \le 40$, `kfree_analytic.py` finds N_k by checking cells of ratio 1.01. For $k > 40$ and $N = \lambda q$ with $\lambda \ge 60$, the requirement reduces to $\lambda(1 - 10^{-11}) - 1 > 4\pi(25.01 \cdot \lambda^{0.05}) + 4$. This holds because the right side is at most 52 for $\lambda \le 10^4$, and grows only like $\lambda^{0.05}$. ■

Proof of part 2. Below N_k the vertex set $V_k \cap [N_k]$ can be listed directly, as the roots of $x^2 \equiv -1 \pmod{p^k}$ for the few primes with $p^k \le N_k^2 + 1$. Testing an edge only needs those same primes. The vertex set is small: 1,143 vertices for $k = 4$, 432 for $k = 5$, and between 90 and 155 for $9 \le k \le 100$. Above $k = 30$ there are no vertices outside the two classes at all.

For $5 \le k \le 100$, `kfree_exact.py` enumerates all maximum cliques at every vertex value up to N_k; the graph is constant between consecutive vertices. For $k = 4$ it does this up to 30,000 and then checks the one-witness bound on 14 cells up to N_4, where the bound is at most a third of c_4. The case $k = 3$ is Theorem 4.1.

The graphs are not trivial: for $k = 5$ there are 844 edges between the two classes and 5 extra vertices. An independent brute force, factorising $x^2 + 1$ and $ab + 1$ directly, reproduces every maximum set for

$k = 4$ up to 3000 and $k = 5$ up to 20,000. ■

So the extremal answer is the same for every $k \geq 3$: the class $r_k \bmod 5^k$, with occasional small exceptions at the start. Part 2 cannot be checked for all k at once. Whether a stray vertex such as 239 falls below the second element of the class depends on the particular roots.

The exceptions are not confined to $c_k(N) \leq 1$: for $k = 3$ they occur where $c_3(N) = 2$. But they do not recur. Let $\mu_k(N)$ be the size of the largest k-valid set in $[N]$ not inside one class. For every $5 \leq k \leq 300$ and every $N \leq N_k$, $\mu_k(N) \leq c_k(N) - 3$. The closest approach is $k = 5$ and $N = 13{,}568$, where a mixed pair faces a class of 5 (`margin.py`). For $12 \leq k \leq 300$ there is no mixed set at all below N_k: $V_k \cap [N_k]$ consists of the two classes, with no edge between them.

5. The forms ab + c

Fix a squarefree $c \geq 1$, and call $A \subseteq [N]$ *c-valid* if $ab + c$ is not squarefree for all $a, b \in A$, including $a = b$. A *root class* is a class $x \bmod p^2$ with $x^2 + c \equiv 0 \pmod{p^2}$. It is self-forcing, because $aa' + c \equiv x^2 + c \equiv 0$ for a, a' in the class.

For $p = 2$, root classes exist exactly when $c \equiv 3 \pmod 4$: they are 1 and 3 mod 4. For odd $p \nmid c$, they exist exactly when $-c$ is a square mod p, and then there are two classes mod p^2. For $p \mid c$ there are none. Let p_c be the least prime with root classes; for $c = 1$ it is 5. The root classes mod p_c^2 are the natural candidates for the maximum. They have density $1/4$ if $p_c = 2$ and $1/p_c^2$ otherwise.

The game for c. The types are built as follows.

- The layer primes are 2, 3 and P, where $P = p_c$ if $p_c \geq 7$ and $P = 5$ otherwise. A type records a mod 36 and a mod P^2.
- Every other prime is a luck prime. For each luck prime $q \mid c$, the type records whether $q \mid a$, as Lemmas 2.2 and 2.4 require.
- The type records membership in the root classes of the least luck prime p with $\rho_p = 2$, as in Section 3.

The constants b_t are those of Lemmas 2.2 and 2.4: union bounds over all luck primes, with the $q \mid c$ terms at their worst case for self-forcing types. Lemma 2.5 then reduces the problem to a finite maximisation, which `game_gen.py` solves with CP-SAT in two steps.

1. It finds the maximum-weight cliques among the types. These are the largest congruence-defined c-valid sets.
2. It maximises $U(K)$ over the cores whose $F(K)$ is not inside any maximum clique. Call this maximum V_c.

Theorem 5.1 (computer-assisted). Let $c \leq 100$ be squarefree with $p_c \leq 7$ and $c \notin \{10, 13\}$; there are 49 such c. Then:

1. The maximum cliques of the game are exactly the two root classes mod p_c^2 (mod 4 if $p_c = 2$).
2. $V_c < 1/p_c^2$ ($V_c < 1/4$ if $p_c = 2$).
3. For all sufficiently large N, the largest c-valid sets in $[N]$ are the root classes mod p_c^2 of largest size, and only these.

Proof. Parts 1 and 2 are the computation. For part 3, let A be c-valid and not inside a root class mod p_c^2. Then the type set of A is not inside a maximum clique, so neither is $F(K)$. By Lemma 2.5, $|A| \leq (V_c + o(1))N$. Since V_c is strictly below the density of the root classes, this is smaller than the

root classes once N is large. A c-valid set inside a root class has at most its size, with equality only for the full class. ■

For 33 of these c, the best congruence-defined c-valid set not inside a root class has density exactly V_c. So V_c is the second-best density.

max density	second-best density	c
1/4 (classes mod 4)	1/9	$3, 11, 23, 47, 51, 59, 71, 83, 87, 95$
1/4	1/18	$43, 67, 79$
1/9 (classes mod 9)	1/12	$2, 14, 26, 38, 62, 74, 86$
1/9	1/18	$5, 17, 29, 41, 53, 65, 89$
1/25 (classes mod 25)	1/50	1
1/25	3/100	$6, 34, 46$
1/49 (classes mod 49)	3/196	82
1/49	1/98	97

For the other 16 values of c ($7, 15, 19, 21, 31, 33, 35, 39, 55, 61, 66, 69, 73, 77, 91, 94$), the best congruence set and V_c still differ, so the second-best density is only bracketed (`gen_k1.jsonl`).

The refinements matter. Without the root-class split and the divisibility flags, the game fails for $c = 34, 61, 66, 69$ and 94: for $c = 34$ it gives $V = 0.0427$ against $1/25$. Each of these has a luck prime dividing c, such as $17 \mid 34$, or many small primes p with $\rho_p = 2$. With the refinements all five pass. With $c = 1$ the computation reproduces $V = 1/50$ of [M].

Exact clique computations (`brute_c.py`) agree with the root class for every N up to 400–800 for $c = 2, 3, 6, 10, 13, 21$ and 34.

Two groups of $c \le 100$ are left open.

- $c = 10$ **and** 13**.** The class is mod 49, and the game with one split prime falls just short: $V_{13} = 0.020465$ against $1/49 = 0.020408$. A second split prime should settle them, but it exceeds our computing budget.
- **The ten** $c \le 100$ **with** $p_c \ge 11$**:** $22, 30, 37, 42, 57, 58, 70, 78, 85$ and 93. The model has p_c^2 slabs, and we have not run it.

Theorem 5.1 gives no explicit threshold for N, since the $o(N)$ terms of Section 2 are not made explicit.

6. Open problems

1. **The third density.** Prove Conjecture 3.2. This needs a two-sided bound for pairs X, Y with $xy + 1$ never squarefree that improves on one witness per side.
2. **All** k**.** Does Theorem 4.3(2) hold for every k? For large k, a mixed set below N_k needs 13^k or 17^k to divide $x^2 + 1$ or $ab + 1$ with x, a, b below about $60 \cdot 5^k$. Heuristically this happens for only finitely many k.
3. **All** c**.** Is the root class mod p_c^2 extremal for every squarefree c, including $c = 10$, 13 and those with $p_c \ge 11$? Is there a proof that does not need a computation for each c? Non-squarefree c brings new self-forcing classes, such as 0 mod q when $q^2 \mid c$, and is not covered here.
4. **Effective thresholds** for Theorems 3.1 and 5.1.

Appendix: reproducibility

All code is in the repository [R], in the folders `third_density/`, `cubefree/`, `kfree/` and `general_c/`. The CP-SAT computations use OR-tools 9.15.

Claim	Computation	Result
Theorem 3.1(1)	`lam3_lower.py`, exact count	0.014671 at $N = 2 \cdot 10^7$
Theorem 3.1(2)	`third_cp.py 13,17 product`	$\max U = 0.0149093$, OPTIMAL; 0.0149645 with S_{13} only; 0.0186 unrefined (`third.py`)
Theorem 4.1	`cube_maxsets.py` ($N \le 700$), `cube_small.py` (to 40,000), `cubeM.c` ($700 \le N \le 2 \cdot 10^6$), `cube_analytic.py` ($N \ge 1.94 \cdot 10^6$)	as stated
Theorem 4.3	`kfree_analytic.py`, `kfree_exact.py`, `run_all.py`, `run_big.py`, `margin.py`	no exceptions for $5 \le k \le 100$; margin at least 3 for $5 \le k \le 300$
Theorem 5.1	`game_gen.py 1 c` for each c; output `gen_k1.jsonl`	$V_c < 1/p_c^2$ for 49 values; $V_1 = 1/50$
Small-N checks for $ab + c$	`brute_c.py`	root class maximal for $c = 2, 3, 6, 10, 13, 21, 34$ up to $N = 400$–800

References

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[RS] J. B. Rosser and L. Schoenfeld, Approximate formulas for some functions of prime numbers, Illinois J. Math. 6 (1962), 64–94.

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